

Regression

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UBC

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Your life in the next 3-4 years

```
. reg price mpg, r
```

Linear regression

```
Number of obs   =      74
F(1, 72)        =     17.28
Prob > F        =     0.0001
R-squared       =     0.2196
Root MSE       =    2623.7
```

price	Robust					
	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
mpg	-238.8943	57.47701	-4.16	0.000	-353.4727	-124.316
_cons	11253.06	1376.393	8.18	0.000	8509.272	13996.85

Data is chaotic, need to make sense of this chaos



Regression must become your security blanket



Regression as Reduction

Regression as Reduction

In order to arrive at a distinct formulation of statistical problems, it is necessary to define the task which the statistician sets himself: briefly, and in its most concrete form, the object of statistical methods is the reduction of data. A quantity of data, which usually by its mere bulk is incapable of entering the mind, is to be replaced by relatively few quantities which shall adequately represent the whole, or which, in other words, shall contain as much as possible, ideally the whole, of the relevant information contained in the original data. – R.A Fisher (1922)

Linear model

- Typical justification for OLS is a linear structural equation relating a dependent variable (the regressand) Y_i to a vector of independent variables (regressors) X_i :

$$Y_i = X_i' \beta + u_i$$

- Suppose $E[X_i u_i] = 0$.
- Then $E[X_i Y_i] = E[X_i X_i'] \beta$, so

$$\beta = E[X_i X_i']^{-1} E[X_i Y_i].$$

- This is the population OLS regression coefficient vector.
- Analogy principle $\implies$ estimate β by replacing population moments with sample moments, so $\hat{\beta} = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{N} \sum_i X_i Y_i \right)$.

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- Pop quiz: what do we mean by “linear structural equation relating [...]”?

$$Y_i = X_i' \beta + u_i$$

- We have seen in the previous class when a regression coefficient has a causal interpretation thanks to random assignment.
- Without random assignment, regression coefficient may or may not have a causal interpretation as we shall see.
- Setting causality aside, regression estimates have “mechanical” statistical properties that are important to know that are intimately related with a fundamental statistical object: the CEF.

- Recall the **conditional expectation** of Y_i given X_i :

$$E[Y_i | X_i = x] = \int_{-\infty}^{\infty} y dF_{Y|X}(y | X_i = x)$$

- The conditional expectation function (CEF), $\mu(x) = E[Y_i | X_i = x]$, describes how the conditional mean of Y_i changes as we vary x
- The CEF can be defined for any Y_i and X_i . It has no behavioral/causal content per se.
- Yet, it is often the starting point to establish the existence of “statistical relationships” between variables.

Summarizing a complex world...

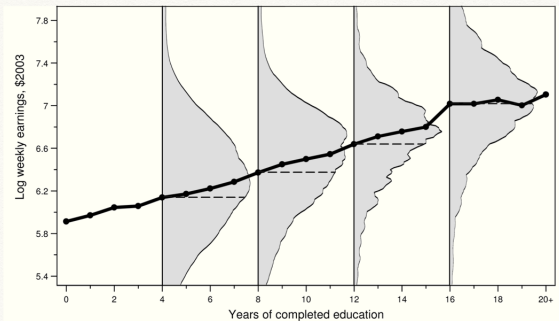


Figure 3.1.1: Raw data and the CEF of average log weekly wages given schooling. The sample includes white men aged 40-49 in the 1980 IPUMS 5 percent file.

CEF Residuals

- We can always write

$$Y_i = \mu(X_i) + u_i$$

- Here $u_i \equiv Y_i - \mu(X_i)$.
- By taking conditional expectations on both sides we obtain

$$E[Y_i|X_i] = \mu(X_i) + E[u_i|X_i]$$

- Since $E[Y_i|X_i] = \mu(X_i)$, we have $E[u_i|X_i] = 0$.
- Note that the CEF residual (or “error term”) u_i has conditional mean zero by definition.
- We have also that
 - 1 $E[u_i] = 0$.
 - 2 $E[h(X_i)u_i] = 0$.

Optimal Predictor

- Suppose we want to use X_i to predict Y_i
- Formally, consider the following minimization problem:

$$\min_{m \in \mathcal{M}} E \left[(Y_i - m(X_i))^2 \right]$$

- Here $\mathcal{M}$ is the set of all measurable functions mapping $\mathcal{R}^K$ to $\mathcal{R}^1$
- This is a complicated problem since we are optimizing over all possible functions

Optimal Predictor

- Rewrite the problem

$$\begin{aligned} & \min_{m \in \mathcal{M}} E \left[([Y_i - \mu(X_i)] + [\mu(X_i) - m(X_i)])^2 \right] \\ &= \min_{m \in \mathcal{M}} E \left[(Y_i - \mu(X_i))^2 \right] + 2E[(Y_i - \mu(X_i))(\mu(X_i) - m(X_i))] \\ & \quad + E[(\mu(X_i) - m(X_i))^2] \end{aligned}$$

- First term doesn't depend on $m(\cdot)$ and the second is zero (why?), so this is equivalent to solving

$$\min_{m \in \mathcal{M}} E[(\mu(X_i) - m(X_i))^2]$$

CEF as Optimal Predictor

$$\min_{m \in \mathcal{M}} E \left[(\mu(X_i) - m(X_i))^2 \right]$$

- This is minimized at zero by setting $\mu(X_i) = m(X_i)$.
- In other words, the CEF is the minimum mean squared error (MMSE) predictor of Y_i .

ANOVA - Law of Total Variance

- Law of total variance

$$\text{Var}(Y_i) = \underbrace{\text{Var}[E[Y_i|X_i]]}_{\text{Variability in explained components}} + \underbrace{E[\text{Var}[Y_i|X_i]]}_{\text{Variability in Unexplained Components}} \quad (1)$$

- Important to study inequality (wage inequality, health inequality, etc...).
- Well defined statistical decomposition.

ANOVA - Law of Total Variance

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- Important to study inequality (wage inequality, health inequality, etc...).
- Well defined statistical decomposition.
- Ok, great but who tells me $\mu(X_i)$?

Linear Projections

- Suppose we want to predict Y_i , but limit ourselves to linear functions
- The **linear projection** of Y_i on X_i is $X_i'\beta$, where

$$\beta \equiv \arg \min_b E \left[(Y_i - X_i'b)^2 \right]$$

- First-order conditions: $E [X_i (Y_i - X_i'\beta)] = 0$
- Then

$$\beta = E [X_i X_i']^{-1} E [X_i Y_i]$$

- This is the OLS coefficient vector
- The OLS projection $X_i'\beta$ is the MMSE **linear** predictor of Y_i .
- We define $L[Y|X] = X_i'\beta$.

Linear Projections and the CEF

- By applying the same trick we saw with the CEF, we can rewrite

$$\begin{aligned}\beta &\equiv \arg \min_b E \left[(Y_i - X_i' b)^2 \right] \\ &= \arg \min_b E \left[(\mu(X_i) - X_i' b)^2 \right]\end{aligned}$$

- OLS therefore also gives the MMSE linear prediction of the **CEF** of Y_i given X_i
- Note: This also implies that when the CEF is linear, OLS fits it exactly

BLP $\rightarrow$ Linear approximation of the CEF

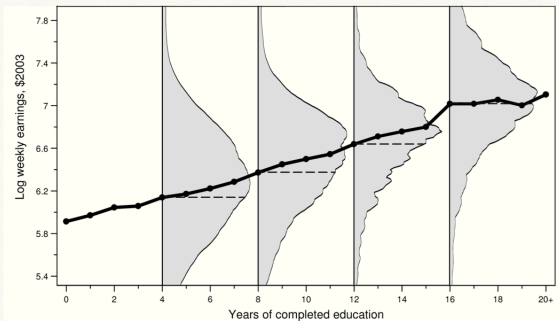


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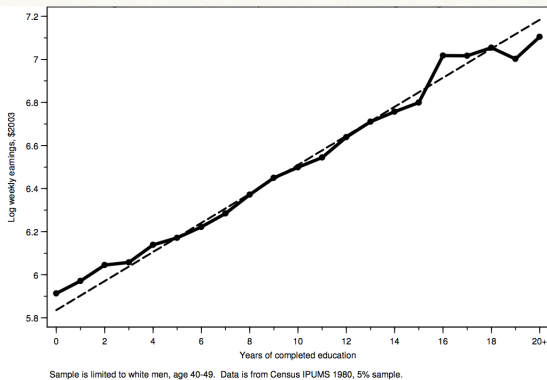


Figure 3.1.2: Regression threads the CEF of average weekly wages given schooling

- Projection model:

$$Y_i = X_i' \beta + U_i$$

- Here $U_i \equiv Y_i - X_i' \beta$ is defined as the difference between Y_i and the least squares projection
- Note that

$$\begin{aligned} E[X_i U_i] &= E[X_i (Y_i - X_i' \beta)] \\ &= 0. \end{aligned}$$

- Note: the projection errors U_i are defined so that X_i and U_i are uncorrelated. They have no “life of their own”.

Omitted Variable Bias

- Consider two regression models:

$$Y_i = \alpha_s + \beta_s S_i + u_{is}$$

$$Y_i = \alpha_\ell + \beta_\ell S_i + \gamma_\ell A_i + u_{i\ell}$$

- Example: “Short” regression of log earnings on schooling, vs. “long” regression of log earnings on schooling and ability (SAT score)

Omitted Variable Bias

$$Y_i = \alpha_s + \beta_s S_i + u_{is}$$

$$Y_i = \alpha_\ell + \beta_\ell S_i + \gamma_\ell A_i + u_{i\ell}$$

- We might prefer the long regression coefficient β_ℓ
- Perhaps higher-ability people go to school longer, but schooling is as good as randomly assigned among people with the same ability
- In other words, the CEF of Y_i given (S_i, A_i) might be more economically interesting than the CEF of Y_i given S_i

Omitted Variables Bias

- Short regression coefficient:

$$\begin{aligned}\beta_s &= \frac{\text{Cov}(Y_i, S_i)}{\text{Var}(S_i)} \\ &= \frac{\text{Cov}(\alpha_\ell + \beta_\ell S_i + \gamma_\ell A_i + u_{i\ell}, S_i)}{\text{Var}(S_i)} \\ &= \beta_\ell + \gamma_\ell \frac{\text{Cov}(A_i, S_i)}{\text{Var}(S_i)}\end{aligned}$$

- The short coefficient equals the long coefficient, plus the coefficient on the omitted variable times the regression of the omitted variable on the included variable.
- This is the standard “omitted variables bias” (OVB) formula. It shows how OLS coefficients change when more regressors are added.
- The long regression coefficient is not necessarily better. Again, it depends on which CEF is of interest.

Good, Bad and Ugly Controls



Good or Bad?

- You want to study whether women are systematically discriminated in the labor market.
- Regression of earnings on gender using age and race as controls returns a coefficient of -10,000 dollars.
- But then you run the same regression also controlling for occupation. Coefficient on gender dummy shrinks to zero.
- Is occupation a good control variable in this context?

Good, Bad and Ugly Controls

- Good controls:
 - Within each value of control, the treatment is as good as randomly assigned.
 - Fixed at the time in which treatment was assigned.
- Bad controls: more is not always better!
 - Variables that should be considered as outcomes per se!
 - Contaminates interpretation of the treatment.
- Ugly controls: what most of us ended up dealing with
 - Need to argue why they are good controls in observational data.
 - Lots of them can create statistical/inference problems → what variation drives my “treatment” (next classes + FWL)?

- Let's now think about how to obtain an estimate of β from a single sample.

$$Y_i = X_i' \beta + u_i \quad (2)$$

- Analogy principle!

$$\hat{\beta} = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{N} \sum_i X_i Y_i \right) \quad (3)$$

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- We will now think about some key algebraic properties of the LS solution (finite sample perspective).

Projection Matrices

- Define the projection matrix:

$$P_X = X(X'X)^{-1}X'$$

- The OLS predicted values are $\hat{y} = X\hat{\beta} = P_X y$.
- P_X projects any vector into the column space of X .
- Likewise, $M_X = I - P_X$ projects any vector into the orthogonal complement. The OLS residual is $\hat{u} = M_X y$.
- P_X is called the “hat” matrix since it “puts a hat” on y , while M_X is the “residual maker” since it turns y into a residual orthogonal to the columns of X

Projection Matrices are awesome! Pt.1

- The diagonal element of the matrix P_X is sometimes called the statistical leverage, call the i th element P_{ii} .
- Important property: $P_{ii} \in [0; 1]$.
- The statistical leverage measures how influential a given observation i in deriving $\hat{\beta}$.

$$P_{ii} = \frac{\partial \hat{y}_i}{\partial y_i} \quad (4)$$

- “Balanced design” means that all observations have equal P_{ii} .
- How influential is observation i for my OLS coefficients?

$$\begin{aligned} \hat{\beta}_{-i} &= (\sum X_{-i} X'_{-i})^{-1} (\sum X_{-i} Y_{-i}) \\ &= \hat{\beta} - (\sum X_i X'_i)^{-1} \left(\sum X_i \frac{\hat{u}_i}{(1 - P_{ii})} \right) \end{aligned} \quad (5)$$

$P_{ij} = 1 \rightarrow$ Overfitting



An Automatic Finite-Sample Robustness Metric: When Can Dropping a Little Data Make a Big Difference?

Tamara Broderick, Ryan Giordano*, and Rachael Meager^{†‡}

July 20, 2023

Abstract

Study samples often differ from the target populations of inference and policy decisions in non-random ways. Researchers typically believe that such departures from random sampling — due to changes in the population over time and space, or difficulties in sampling truly randomly — are small, and their corresponding impact on the inference should be small as well. We might therefore be concerned if the conclusions of our studies are excessively sensitive to a very small proportion of our sample data. We propose a method to assess the sensitivity of applied econometric conclusions to the removal of a small fraction of the sample. Manually checking the influence of all possible small subsets is computationally infeasible, so we use an approximation to find the most influential subset. Our metric, the “Approximate Maximum Influence Perturbation,” is based on the classical influence function, and is automatically computable for common methods including (but not limited to) OLS, IV, MLE, GMM, and variational Bayes. We provide finite-sample error bounds on approximation performance. At minimal extra cost, we provide an exact finite-sample lower bound on sensitivity. We find that sensitivity is driven by a signal-to-noise ratio in the inference problem, is not reflected in standard errors, does not disappear asymptotically, and is not due to misspecification. While some empirical applications are robust, results of several influential economics papers can be overturned by removing less than 1% of the sample.

Projection Matrices are awesome Pt.2

- Useful properties of projection matrices:
 - Symmetric: $P_X = P_X'$, $M_X = M_X'$
 - Idempotent: $P_X P_X = P_X$, $M_X M_X = M_X$
 - Zero product: $P_X M_X = 0$
 - $P_X X = X$, and $M_X X = 0$ (M_X “annihilates” X)

Frisch-Waugh-Lovell Theorem

- Consider the regression model:

$$y = X_1\beta_1 + X_2\beta_2 + u$$

- We can write $y = X_1\hat{\beta}_1 + X_2\hat{\beta}_2 + \hat{u}$, where $\hat{\beta}_1$ and $\hat{\beta}_2$ are OLS estimates and $\hat{u}$ is the residual
- Instead of running this regression, let's first take residuals from regressions of y and X_2 on X_1 :

$$\tilde{y} = M_1y, \tilde{X}_2 = M_1X_2$$

- Then regress $\tilde{y}$ on $\tilde{X}_2$:

$$\tilde{y} = \tilde{X}_2\gamma_2 + u_2$$

Frisch-Waugh-Lovell Theorem

- OLS estimate of γ_2 :

$$\begin{aligned}\hat{\gamma}_2 &= (\tilde{X}_2' \tilde{X}_2)^{-1} \tilde{X}_2' \tilde{y} \\ &= (X_2' M_1 M_1 X_2)^{-1} X_2' M_1 M_1 y \\ &= (X_2' M_1 X_2)^{-1} X_2' M_1 (X_1 \hat{\beta}_1 + X_2 \hat{\beta}_2 + \hat{u}) \\ &= (X_2' M_1 X_2)^{-1} (X_2' M_1 X_1 \hat{\beta}_1 + X_2' M_1 X_2 \hat{\beta}_2 + X_2' M_1 \hat{u})\end{aligned}$$

Frisch-Waugh-Lovell Theorem

$$\hat{\gamma}_2 = (X_2' M_1 X_2)^{-1} (X_2' M_1 X_1 \hat{\beta}_1 + X_2' M_1 X_2 \hat{\beta}_2 + X_2' M_1 \hat{u})$$

- The first term in parentheses is zero
- $M_1 \hat{u} = \hat{u}$ and $X_2' \hat{u} = 0$ so the third term is also zero. Then

$$\begin{aligned}\hat{\gamma}_2 &= (X_2' M_1 X_2)^{-1} X_2' M_1 X_2 \hat{\beta}_2 \\ &= \hat{\beta}_2.\end{aligned}$$

- The “long regression” coefficient on X_2 , $\hat{\beta}_2$, is therefore the same as the “residual regression” coefficient, $\hat{\gamma}_2$. This is the first part of the FWL theorem.

Frisch-Waugh-Lovell Theorem

- The second part of the FWL theorem is obtained by multiplying both sides of our expression for y by M_1 :

$$\begin{aligned}M_1 y &= M_1 X_1 \hat{\beta}_1 + M_1 X_2 \hat{\beta}_2 + M_1 \hat{u} \\ \implies \tilde{y} &= \tilde{X}_2 \hat{\beta}_2 + M_1 \hat{u}\end{aligned}$$

- Since $\hat{\beta}_2$ is the coefficient from the residual regression, $M_1 \hat{u}$ is the residual from this regression. Since $M_1 \hat{u} = \hat{u}$, the residual regression generates the same residuals as the long regression. This is the second part of the FWL theorem.
- This result means that standard errors obtained from the residual regression will also match the long regression (with an appropriate degrees of freedom correction)

Frisch-Waugh-Lovell Theorem: Summary

- Together, the two parts of the FWL theorem say that if we're interested in a subset of the coefficients from a multivariate regression, we can first “partial out” the other variables from both sides, then work with the residuals.
- This procedure will generate both the same coefficients and the same residuals we'd get from the long regression.
- Other notes:
 - We can also obtain the same coefficients by first partialling out some variables from the right-hand side but not the left. This will not produce the same residuals, however.
 - In general, we will NOT obtain the same coefficients by partialling out variables from the left but not the right.

Analysis of Variance

- Now that we understand the mechanics of regression, let's consider a few ways it's used in practice
- One traditional use, proposed by R.A. Fisher, is “Analysis of Variance” (ANOVA)
- ANOVA simply asks whether the mean of some variable differs across observed groups
- Name derives from fact that in linear model can partition variance of response into components due to “fixed” predictors $X'\beta$ and error.

Analysis of Variance

- Suppose we have data on wages w_i for a sample of individuals
- We also have data on race ($black_i \in \{0, 1\}$) and gender ($female_i \in \{0, 1\}$)
- Suppose we run the regression

$$w_i = \alpha_0 + \alpha_1 black_i + \alpha_2 female_i + \alpha_3 black_i \cdot female_i + u_i$$

- How should we interpret the coefficients $(\alpha_0, \alpha_1, \alpha_2, \alpha_3)$?

Analysis of Variance

$$w_i = \alpha_0 + \alpha_1 \text{black}_i + \alpha_2 \text{female}_i + \alpha_3 \text{black}_i \cdot \text{female}_i + u_i$$

- This is a **saturated** model: The number of coefficients equals the number of possible values of the right-hand-side variables
- In a saturated model the CEF is necessarily linear
- As we've learned, regression perfectly fits the CEF when the CEF is linear

Analysis of Variance

$$w_i = \alpha_0 + \alpha_1 \text{black}_i + \alpha_2 \text{female}_i + \alpha_3 \text{black}_i \cdot \text{female}_i + u_i$$

- Then

$$\alpha_0 = E[w_i | \text{black}_i = 0, \text{female}_i = 0]$$

$$\alpha_1 = E[w_i | \text{black}_i = 1, \text{female}_i = 0] - E[w_i | \text{black}_i = 0, \text{female}_i = 0]$$

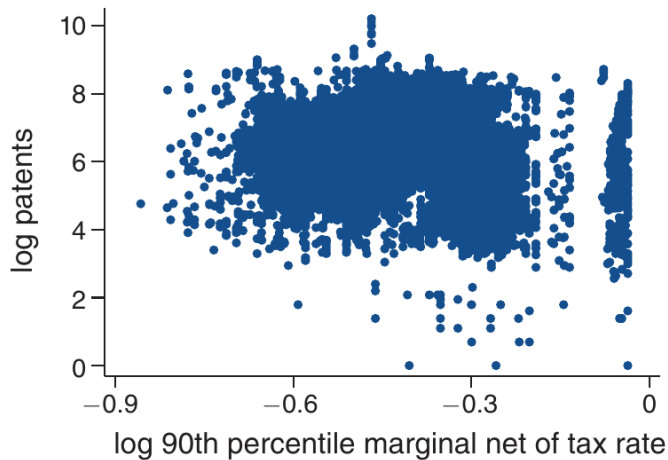
$$\alpha_2 = E[w_i | \text{black}_i = 0, \text{female}_i = 1] - E[w_i | \text{black}_i = 0, \text{female}_i = 0]$$

$$\alpha_3 = (E[w_i | \text{black}_i = 1, \text{female}_i = 1] - E[w_i | \text{black}_i = 1, \text{female}_i = 0]) \\ - (E[w_i | \text{black}_i = 0, \text{female}_i = 1] - E[w_i | \text{black}_i = 0, \text{female}_i = 0])$$

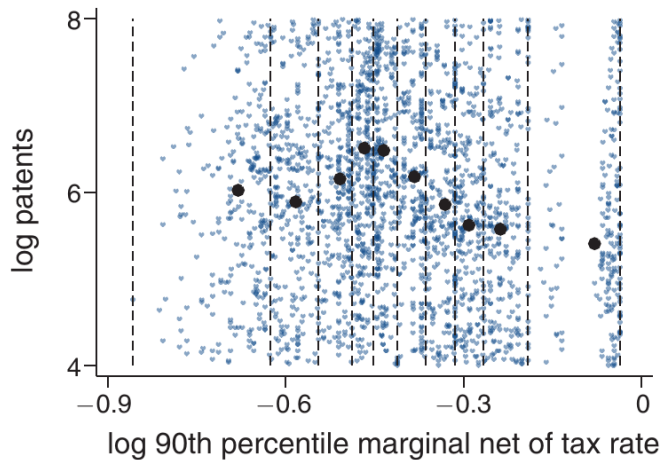
- Any question about differences in means across groups can be answered with these parameters

- We are interested in studying the relationship between Y and X . X is (basically) continuous, thus it is hard to estimate CEF.
- Example: Y_{st} log number of patents in state s and year t . X_{st} is the log 90th percentile marginal net of tax rate (Akcigit et al., 2022).
- Can run a regression but we are worried that the relationship might not be linear (why?).
- What would you do?

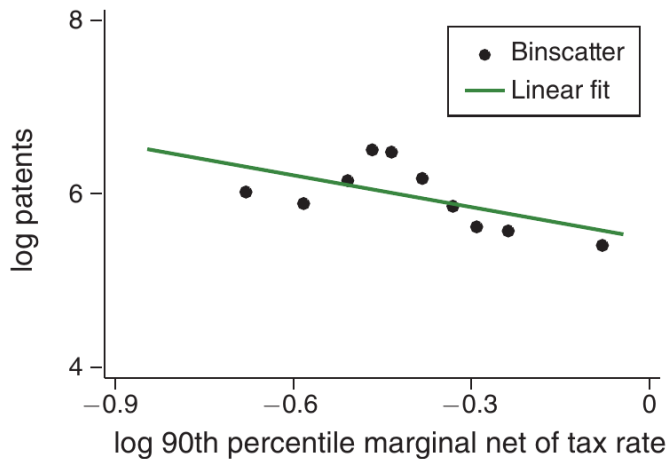
Panel A. Raw scatter plot



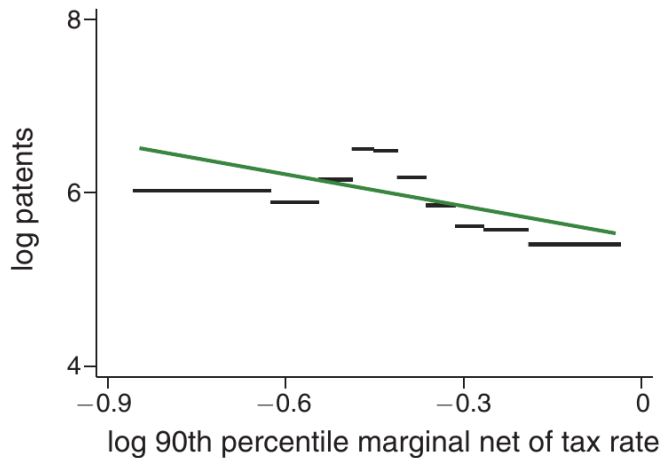
Panel B. Bin cutoffs and binscatter



Panel C. Binscatter plot



Panel D. Conditional mean point est.



Covariate adjustment: do not residualize X

- Standard practice: regress Y and X on W , then binscatter the two residuals. The justification given is FWL.

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- Residualizing is valid only if $\mu(\cdot)$ is linear, which is exactly what we wanted the plot to tell us.

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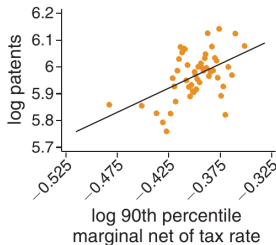
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- Two things break: the horizontal axis is no longer X , and a nonlinear $\mu(\cdot)$ can be drawn as a straight line.
- Fix: keep the bins in X and let W enter the same regression

$$Y_i = \sum_{j=1}^J \beta_j 1\{X_i \in \mathcal{I}_j\} + W_i' \gamma + e_i$$

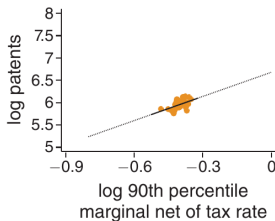
and plot $\hat{\beta}_j$. Partial W out of the bin dummies, not out of X .

The same data, three ways

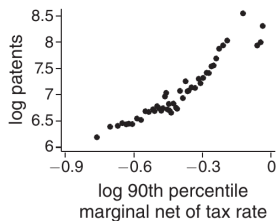
Panel A. Standard binscatter
(incorrect scale)



Panel B. Standard binscatter
(correct scale)



Panel C. Corrected binscatter
(correct scale)



What if we want to plot this relationship after having controlled for some variables?

What if we want to plot this relationship after having controlled for some variables?

On Binscatter[†]

By MATIAS D. CATTANEO, RICHARD K. CRUMP, MAX H. FARRELL,
AND YINGJIE FENG*

Binscatter is a popular method for visualizing bivariate relationships and conducting informal specification testing. We study the properties of this method formally and develop enhanced visualization and econometric binscatter tools. These include estimating conditional means with optimal binning and quantifying uncertainty. We also highlight a methodological problem related to covariate adjustment that can yield incorrect conclusions. We revisit two applications using our methodology and find substantially different results relative to those obtained using prior informal binscatter methods. General purpose software in Python, R, and Stata is provided. Our technical work is of independent interest for the nonparametric partition-based estimation literature. (JEL C13, C14, C18, C51, O31, R32)

Getting Practical

- Lots of textbooks (1st grad econometrics) might have told you that running a regression is fine only if the outcome is “continuous” (e.g. log wage).

Getting Practical

- Lots of textbooks (1st grad econometrics) might have told you that running a regression is fine only if the outcome is “continuous” (e.g. log wage).
- This recommendation is misguided and you shall not follow it.
- If the interest lies in assessing the (average) causal impact of some treatment D on outcome Y , running a regression of Y on D will return the ATE when there is random assignment of D , regardless of whether Y is continuous, discrete, binary.
- The same does not hold if running a probit regression on Y on D (and you have random assignment of D)!.

Getting Practical

- Often you encounter an outcome variable where for lots of observations are such that $Y_i = 0$.

Getting Practical

- Often you encounter an outcome variable where for lots of observations are such that $Y_i = 0$.
- Provided that $Y_i > 0$, Y_i then looks like a continuous variable.
 - 1 Provide me an example of such variable.
 - 2 A lot of people in these situations run a regression of $\log(Y_i)$ on D_i . Is that regression identifying the causal effect of D_i on Y_i ?
 - 3 Other people instead run a regression of $\log(1 + Y_i)$ on D_i ? What do you think is the rationale behind that?

LOGS WITH ZEROS? SOME PROBLEMS AND SOLUTIONS*

JIAFENG CHEN AND JONATHAN ROTH

When studying an outcome Y that is weakly positive but can equal zero (e.g., earnings), researchers frequently estimate an average treatment effect (ATE) for a “log-like” transformation that behaves like $\log(Y)$ for large Y but is defined at zero (e.g., $\log(1 + Y)$, $\operatorname{arcsinh}(Y)$). We argue that ATEs for log-like transformations should not be interpreted as approximating percentage effects, since unlike a percentage, they depend on the units of the outcome. In fact, we show that if the treatment affects the extensive margin, one can obtain a treatment effect of any magnitude simply by rescaling the units of Y before taking the log-like transformation. This arbitrary unit dependence arises because an individual-level percentage effect is not well-defined for individuals whose outcome changes from zero to nonzero when receiving treatment, and the units of the outcome implicitly determine how much weight the ATE for a log-like transformation places on the extensive margin. We further establish a trilemma: when the outcome can equal zero, there is no treatment effect parameter that is an average of individual-level treatment effects, unit invariant, and point identified. We discuss several alternative approaches that may be sensible in settings with an intensive and extensive margin, including (i) expressing the ATE in levels as a percentage (e.g., using Poisson regression), (ii) explicitly calibrating the value placed on the intensive and extensive margins, and (iii) estimating separate effects for the two margins (e.g., using Lee bounds). We illustrate these approaches in three empirical applications. *JEL codes:* C01.

Connecting Regression with Treatment Effects

- Suppose we run randomized experiments at several sites
- Treatment $T_i \in \{0, 1\}$ is randomly assigned within a site, but the probability of assignment differs across sites
- We therefore only want to compare treatment and control groups within sites
- Let $X_i \in \{1 \dots K\}$ denote site, and let $D_i = (1 \{X_i = 1\}, \dots, 1 \{X_i = K\})'$ denote a vector of site indicators

Stratified Experiments

- We run the OLS regression:

$$Y_i = D_i' \Pi + \delta T_i + u_i$$

- This model includes a dummy for every site, but only a main effect of the treatment → Is this regression saturated?
- What does δ identify in terms of potential outcomes (Y_{i1}, Y_{i0}) ?

Stratified Experiments

- We run the OLS regression:

$$Y_i = D_i' \Pi + \delta T_i + u_i$$

- This model includes a dummy for every site, but only a main effect of the treatment $\rightarrow$ Is this regression saturated?
- What does δ identify in terms of potential outcomes (Y_{i1}, Y_{i0}) ?
- Recall that we can always write

$$Y_i = T_i Y_{i1} + (1 - T_i) Y_{i0} \quad (6)$$

Stratified Experiments

$$Y_i = D_i' \Pi + \delta T_i + u_i$$

- Let's use our knowledge of the FWL theorem to derive an expression for δ :

$$\delta = E [\tilde{T}_i^2]^{-1} E [\tilde{T}_i \tilde{Y}_i]$$

- $\tilde{T}_i$ and $\tilde{Y}_i$ are residuals from regressions on D_i
- We can rewrite δ as follows... why?

$$\delta = \frac{E [(Y_i - E[Y_i|X_i]) (T_i - E[T_i|X_i])]}{E [(T_i - E[T_i|X_i])^2]}$$

Stratified Experiments

$$\delta = \frac{E[Y_i(T_i - E[T_i|X_i])]}{E[(T_i - E[T_i|X_i])^2]} - \frac{E[E[Y_i|X_i](T_i - E[T_i|X_i])]}{E[(T_i - E[T_i|X_i])^2]}$$

- The second term is zero (why?)
- By iterated expectations the first term can be written

$$\delta = \frac{E[E[Y_i|X_i, T_i](T_i - E[T_i|X_i])]}{E[(T_i - E[T_i|X_i])^2]}$$

Stratified Experiments

- It's useful to rewrite

$$E[Y_i|X_i, T_i] = E[Y_i|X_i, T_i = 0] + \Delta(X_i)T_i,$$

$$\Delta(X_i) = E[Y_i|X_i, T_i = 1] - E[Y_i|X_i, T_i = 0]$$

- Here $\Delta(X_i)$ is a site-specific treatment/control contrast.
- We can then write (why?):

$$E[Y_i|X_i, T_i = 0] = E[Y_{i0}|X_i] \tag{7}$$

$$\Delta(X_i) = E[Y_{i1} - Y_{i0}|X_i] \tag{8}$$

- Putting pieces together now...

$$\begin{aligned}
 \delta &= \frac{E[(E[Y_i|X_i, T_i = 0] + \Delta(X_i)T_i)(T_i - E[T_i|X_i])]}{E[(T_i - E[T_i|X_i])^2]} \\
 &= \frac{E[\Delta(X_i)T_i(T_i - E[T_i|X_i])]}{E[(T_i - E[T_i|X_i])^2]} \\
 &= \frac{E[E[Y_{i1} - Y_{i0}|X_i]T_i(T_i - E[T_i|X_i])]}{E[(T_i - E[T_i|X_i])^2]}
 \end{aligned}$$

Stratified Experiments

- By invoking iterated expectations one more time we obtain

$$\delta = \frac{E[E[Y_{i1} - Y_{i0}|X_i]E[T_i(T_i - E[T_i|X_i])|X_i]]}{E[E[(T_i - E[T_i|X_i])^2|X_i]]}$$

Stratified Experiments

- By invoking iterated expectations one more time we obtain

$$\begin{aligned}\delta &= \frac{E[E[Y_{i1} - Y_{i0}|X_i]E[T_i(T_i - E[T_i|X_i])|X_i]]}{E[E[(T_i - E[T_i|X_i])^2|X_i]]} \\&= \frac{E[E[Y_{i1} - Y_{i0}|X_i]\sigma_T^2(X_i)]}{E[\sigma_T^2(X_i)]} \\&= E[E[Y_{i1} - Y_{i0}|X_i]w(X_i)]\end{aligned}$$

- Here $w(X_i) = \sigma_T^2(X_i) / E[\sigma_T^2(X_i)]$
- Since treatment is binary $\sigma_T^2(X_i) = \text{Pr}[T_i = 1|X_i](1 - \text{Pr}[T_i = 1|X_i])$
- OLS with controls for site indicators therefore identifies a variance-of-treatment-weighted average of site-specific treatment/control comparisons
- Works even in the contest of misspecification of the CEF!
- Useful statistical object even when T_i is not randomly assigned within groups

Going one step beyond

New scenario: two treatments are now randomized within the site

Run the following regression

$$Y_i = D_i\Pi + \delta_1 T_{i1} + \delta_2 T_{i2} + u_i \quad (9)$$

- Randomization implies $(Y_{i0}, Y_{i1}, Y_{i2}) \perp (T_{i1}, T_{i2}) | D_i$

What is (δ_1, δ_2) identifying?

Contamination Bias in Linear Regressions*

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June 2, 2022

Abstract

We study the interpretation of regressions with multiple treatments and flexible controls. Such regressions are often used to analyze stratified randomized control trials with multiple intervention arms, to estimate value-added (for, e.g., teachers) with observational data, and to leverage the quasi-random assignment of decision-makers (e.g. bail judges). We show that these regressions generally fail to estimate convex averages of heterogeneous treatment effects, even when the treatments are conditionally randomly assigned and the controls are sufficiently flexible to avoid omitted variables bias. Instead, estimates of each treatment's effects are generally contaminated by a non-convex average of the effects of other treatments. Thus, recent concerns about heterogeneity-induced bias in regressions leveraging potential outcome restrictions (e.g. parallel trends assumptions) also arise with “design-based” identification strategies. We discuss solutions to the contamination bias and propose a new class of efficient estimators of weighted average effects that avoid bias. In a re-analysis of the Project STAR trial, we find minimal bias because treatment effect heterogeneity is largely idiosyncratic. But sizeable contamination bias arises when effect heterogeneity becomes correlated with treatment propensity scores.

Where is the failure?

Apply FWL again, now residualizing T_{i1} on both D_i and T_{i2}

$$\delta_1 = E \left[\tilde{T}_{i1}^2 \right]^{-1} E \left[\tilde{T}_{i1} Y_i \right], \quad \tilde{T}_{i1} \equiv T_{i1} - L \left[T_{i1} | X_i, T_{i2} \right]$$

- (No need to residualize Y_i —why?)

The observed outcome can be expressed in terms of 3 POs

$$Y_i = Y_{i0} + \underbrace{(Y_{i1} - Y_{i0})}_{\tau_{i1}} T_{i1} + \underbrace{(Y_{i2} - Y_{i0})}_{\tau_{i2}} T_{i2} \quad (10)$$

Substituting splits δ_1 into three pieces

$$\delta_1 = \underbrace{\frac{E \left[Y_{i0} \tilde{T}_{i1} \right]}{E \left[\tilde{T}_{i1}^2 \right]}}_{(I)} + \underbrace{\frac{E \left[\tau_{i1} T_{i1} \tilde{T}_{i1} \right]}{E \left[\tilde{T}_{i1}^2 \right]}}_{(II)} + \underbrace{\frac{E \left[\tau_{i2} T_{i2} \tilde{T}_{i1} \right]}{E \left[\tilde{T}_{i1}^2 \right]}}_{(III)}$$

- III has no counterpart in the one-treatment case: the other arm's effect loading onto this arm

What we residualize on now

Notation: $p_k(X_i) = \Pr[T_{ik}=1|X_i]$ for $k = 0, 1, 2$

With one treatment, saturating D_i made the projection the CEF

$$L[T_i|X_i] = E[T_i|X_i] = p(X_i)$$

With two, the regression forces

$$L[T_{i1}|X_i, T_{i2}] = a(X_i) + b T_{i2}$$

- One free intercept $a(x)$ per site, so still saturated in X_i
- Single slope b on T_{i2}
- The true CEF $E[T_{i1}|X_i, T_{i2}]$ is not of this form

This misspecification is the source of all the issues in this regression!

Solving for $a(\cdot)$

Saturation in X_i buys one first-order condition per site

$$E [\tilde{T}_{i1} \mathbf{1}[X_i=x]] = 0 \quad \implies \quad E [\tilde{T}_{i1} | X_i] = 0$$

Apply it to $\tilde{T}_{i1} = T_{i1} - a(X_i) - b T_{i2}$

$$0 = p_1(X_i) - a(X_i) - b p_2(X_i) \quad \implies \quad a(x) = p_1(x) - b p_2(x)$$

Substituting back, the residual is a difference of within-site demeaned indicators,

$$\dot{T}_{ik} \equiv T_{ik} - p_k(X_i)$$

$$\tilde{T}_{i1} = \dot{T}_{i1} - b \dot{T}_{i2}$$

Solving for b

One condition is left. Sub in $\tilde{T}_{i1} = \dot{T}_{i1} - b \dot{T}_{i2}$ and $T_{i2} = \dot{T}_{i2} + p_2(X_i)$

$$\begin{aligned} 0 &= E [\tilde{T}_{i1} T_{i2}] = E [(\dot{T}_{i1} - b \dot{T}_{i2}) (\dot{T}_{i2} + p_2(X_i))] \\ &= E [(\dot{T}_{i1} - b \dot{T}_{i2}) \dot{T}_{i2}] + E [(\dot{T}_{i1} - b \dot{T}_{i2}) p_2(X_i)] \\ &= E [\dot{T}_{i1} \dot{T}_{i2}] - b E [\dot{T}_{i2}^2] \end{aligned}$$

so that

$$b = \frac{E [\dot{T}_{i1} \dot{T}_{i2}]}{E [\dot{T}_{i2}^2]}$$

Solving for b

The conditional moments follow from $T_{i1} T_{i2} = 0$

$$E[\dot{T}_{i1} \dot{T}_{i2} | X_i] = -p_1(X_i)p_2(X_i), \quad E[\dot{T}_{ik}^2 | X_i] = p_k(X_i)(1 - p_k(X_i))$$

Writing $\sigma_2^2(x) = p_2(x)(1 - p_2(x)) = \text{Var}[T_{i2} | X_i = x]$

$$b = \frac{-E[p_1(X_i)p_2(X_i)]}{E[\sigma_2^2(X_i)]} < 0$$

- The arms are mechanically negatively correlated within site
- Residualizing on T_{i2} does something even under random assignment

The CEF we should have used

Because the arms are mutually exclusive, the true CEF is

$$E[T_{i1}|X_i, T_{i2}] = (1 - T_{i2}) \frac{p_1(X_i)}{1 - p_2(X_i)} = \underbrace{\frac{p_1(X_i)}{1 - p_2(X_i)}}_{a^*(X_i)} + \underbrace{\frac{-p_1(X_i)}{1 - p_2(X_i)}}_{b(X_i)} T_{i2}$$

- T_{i2} is binary, so this is exactly linear in T_{i2} , with no approximation

Its slope is

$$\begin{aligned} b(x) &= \frac{-p_1(x)}{1 - p_2(x)} = -\Pr[T_{i1}=1|X_i=x, T_{i2}=0] \in [-1, 0] \\ &= \frac{-p_1(x)p_2(x)}{\sigma_2^2(x)} \end{aligned}$$

b versus $b(x)$

Since $b(x)\sigma_2^2(x) = -p_1(x)p_2(x)$, the single b we estimate is

$$b = \frac{E[b(X_i)\sigma_2^2(X_i)]}{E[\sigma_2^2(X_i)]}$$

- A variance-weighted average of the site-specific slopes
- As a convex combination of the $b(x)$, $b \in [-1, 0]$ too

When will the regression be correctly specified?

- The misspecification in this slope, $b(x) - b$, will cause bias in δ_1

Taking the three terms in turn

Recall the decomposition

$$\delta_1 = \underbrace{\frac{E[Y_{i0} \tilde{T}_{i1}]}{E[\tilde{T}_{i1}^2]}}_{(I)} + \underbrace{\frac{E[\tau_{i1} T_{i1} \tilde{T}_{i1}]}{E[\tilde{T}_{i1}^2]}}_{(II)} + \underbrace{\frac{E[\tau_{i2} T_{i2} \tilde{T}_{i1}]}{E[\tilde{T}_{i1}^2]}}_{(III)}$$

and write $\Delta_k(X_i) = E[\tau_{ik}|X_i] = E[Y_{ik} - Y_{i0}|X_i]$ for the site-level effect of arm k

- (I) Selection goes to zero
- (II) Own effect is a convex but distorted weighted average of the $\Delta_1(x)$
- (III) Contamination is arm-2 effects loading onto δ_1 , with weights that are not sign-restricted (!)

Term (I): selection

Our friend, the law of iterated expectations

$$E[Y_{i0} \tilde{T}_{i1}] = E \left[\underbrace{E[Y_{i0}|X_i]}_{\text{selection on } X} \cdot \underbrace{E[\tilde{T}_{i1}|X_i]}_{=0 \text{ by saturation}} \right] = 0$$

Why is $\tilde{T}_{i1} \perp Y_{i0}|X$ and why does $E[\tilde{T}_{i1}|X_i] = 0$?

- Surprising, because it is linearly residualized?
- (check the derivation of $a(x)$!)
- The misspecification surfaces in (II) and (III)

Term (II): the numerator

Random assignment lets us condition down to the site level

$$E [\tau_{i1} T_{i1} \tilde{T}_{i1}] = E [\Delta_1(X_i) \omega(X_i)], \quad \omega(X_i) \equiv E [T_{i1} \tilde{T}_{i1} | X_i]$$

Using $\tilde{T}_{i1} = \dot{T}_{i1} - b \dot{T}_{i2}$ and $T_{i1} T_{i2} = 0$

$$E[T_{i1} \dot{T}_{i1} | X_i] = p_1(1 - p_1), \quad E[T_{i1} \dot{T}_{i2} | X_i] = -p_1 p_2$$

so the numerator is

$$E \left[\Delta_1(X_i) \left(\underbrace{p_1(X_i) (1 - p_1(X_i))}_{\text{the one-treatment weight}} + \underbrace{b p_1(X_i) p_2(X_i)}_{\text{new}} \right) \right]$$

- The new piece includes b , so the weight on site x depends on the slopes at every other site

Term (II): the denominator

Write $V_1 = E[p_1(1 - p_1)]$, $V_2 = E[\sigma_2^2] = E[p_2(1 - p_2)]$ and $C = E[p_1 p_2]$, so that $b = -C/V_2$

$$E[\tilde{T}_{i1}^2] = V_1 + 2bC + b^2 V_2$$

Substitute the definition $b = -C/V_2$ into the last term, $b^2 V_2 = -bC$

$$E[\tilde{T}_{i1}^2] = V_1 - \frac{C^2}{V_2} = V_1 + bC$$

$$= E[p_1(X_i)(1 - p_1(X_i)) + b p_1(X_i)p_2(X_i)] = E[\omega(X_i)]$$

- Exactly the numerator, with Δ_1 stripped out, so the weights self-normalize

Term (II): a weighted average of causal effects

Term (II) is a weighted average of the site-level effects of arm 1

$$(II) = E [\Delta_1(X_i) w(X_i)], \quad w(X_i) = \frac{\omega(X_i)}{E[\omega(X_i)]}, \quad E[w(X_i)] = 1$$

Are the weights positive? Substitute $1 - p_1 = p_2 + p_0$ into ω

$$\omega(x) = p_1(x) [p_0(x) + (1 + b) p_2(x)] \geq 0, \quad \text{since } 1 + b \in [0, 1]$$

- So (II) is a convex combination and cannot push δ_1 outside the range of the $\Delta_1(x)$
- Different weights than the single-treatment setting

What is the intuition for the weights?

- Compare this to the single-treatment setting
- Special case of $b=0$ is particularly instructive

Term (III): contamination

The same steps give

$$E [\tau_{i2} T_{i2} \tilde{T}_{i1}] = E [\Delta_2(X_i) c(X_i)], \quad c(X_i) \equiv E [T_{i2} \tilde{T}_{i1} | X_i]$$

Now $E[T_{i2} \dot{T}_{i1} | X_i] = -p_1 p_2$ and $E[T_{i2} \dot{T}_{i2} | X_i] = \sigma_2^2$

$$c(X_i) = -p_1(X_i)p_2(X_i) - b\sigma_2^2(X_i) = \underbrace{\sigma_2^2(X_i)}_{\text{assignment variance}} \times \underbrace{[b(X_i) - b]}_{\text{prediction error}}$$

- **A variance-weighted prediction error:** how variable arm-2 assignment is at x , times how wrong the common slope is at x

These weights sum to zero

$$E [c(X_i)] = E [\sigma_2^2(X_i) (b(X_i) - b)] = 0$$

The contamination term is a covariance

$$(III) = \frac{\text{Cov}(\Delta_2(X_i), c(X_i))}{E [\tilde{T}_{i1}^2]}$$

Reading the contamination term

Combining the two expressions

$$(III) = \frac{E [\Delta_2(X_i) \sigma_2^2(X_i) (b(X_i) - b)]}{E [\tilde{T}_{i1}^2]}$$

- $c(\cdot)$ is a pure function of the design: assignment shares only, no potential outcomes
- Arm-2 effects load onto δ_1 wherever a site's own slope departs from the average, magnified by how variable arm-2 assignment is there

The weights can be positive or negative

- Sites with $b(x) > b$ contribute their $\Delta_2(x)$ positively
- Sites with $b(x) < b$ contribute negatively
- δ_1 can be negative even when every $\Delta_1(x)$ and $\Delta_2(x)$ is positive (why?)

When does none of this matter?

Constant treatment effects

- If $\Delta_2(x) = \Delta_2$ for all x , then $(III) = \Delta_2 E[c(X_i)] / E[\tilde{T}_{i1}^2] = 0$
- If also $\Delta_1(x) = \Delta_1$, then $(II) = \Delta_1$ since $E[w] = 1$
- Heterogeneity is the whole problem

Constant $b(\cdot)$, so that $c(x) = 0$

- For δ_1 alone this is weaker than equal shares
 - E.g. $(p_1, p_2) = (0.2, 0.5)$ and $(0.3, 0.25)$ both give $b(x) = -0.4$
- But to make δ_2 interpretable, we need $b_2(x) = -p_2(x)/(1 - p_1(x))$ constant too
- Both constant (and $p_0(x) > 0$ somewhere) $\rightarrow$ constant $p_1(\cdot)$ and $p_2(\cdot)$
- Then $\omega(x) = \bar{p}_1 \bar{p}_0 / (\bar{p}_1 + \bar{p}_0)$ is constant, $w \equiv 1$, and $\delta_1 = E[\Delta_1(X_i)] = ATE_1$

What else can we do?

When does none of this matter?

Constant treatment effects

- If $\Delta_2(x) = \Delta_2$ for all x , then $(III) = \Delta_2 E[c(X_i)] / E[\tilde{T}_{i1}^2] = 0$
- If also $\Delta_1(x) = \Delta_1$, then $(II) = \Delta_1$ since $E[w] = 1$
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Constant $b(\cdot)$, so that $c(x) = 0$

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 - E.g. $(p_1, p_2) = (0.2, 0.5)$ and $(0.3, 0.25)$ both give $b(x) = -0.4$
- But to make δ_2 interpretable, we need $b_2(x) = -p_2(x)/(1 - p_1(x))$ constant too
- Both constant (and $p_0(x) > 0$ somewhere) $\rightarrow$ constant $p_1(\cdot)$ and $p_2(\cdot)$
- Then $\omega(x) = \bar{p}_1 \bar{p}_0 / (\bar{p}_1 + \bar{p}_0)$ is constant, $w \equiv 1$, and $\delta_1 = E[\Delta_1(X_i)] = ATE_1$

What else can we do?

- Saturate and weight
- I.e., estimate site-specific effects and aggregate with chosen weights

Table 1: Project STAR Contamination Bias and Treatment Effect Estimates

	A. Contamination Bias Estimates				
	Regression	Own	Bias	Worst-Case Bias	
	Coefficient	Effect		Negative	Positive
	(1)	(2)	(3)	(4)	(5)
Small Class Size	5.357 (0.778)	5.202 (0.778)	0.155 (0.160)	-1.654 (0.185)	1.670 (0.187)
Teaching Aide	0.177 (0.720)	0.360 (0.714)	-0.183 (0.149)	-1.529 (0.176)	1.530 (0.177)
	B. Treatment Effect Estimates				
	Unweighted	Efficiently-Weighted			
	(ATE)	One-at-a-time	Common		
	(1)	(2)	(3)		
Small Class Size	5.561 (0.763) [0.744]	5.295 (0.775) [0.743]	5.563 (0.764) [0.742]		
Teaching Aide	0.070 (0.708) [0.694]	0.263 (0.715) [0.691]	-0.003 (0.712) [0.695]		

Notes: Panel A estimates the contamination bias and range of potential contamination bias in regression estimates of small class and teaching aide treatment effects for the Project STAR kindergarten analysis. The analysis sample includes 5,868 students. Column 1 reports estimates from a partially linear model in eq. (25). Columns 2 and 3 estimate the own- and cross-treatment decomposition of this estimate in eq. (27). Columns 4 and 5 reports the smallest (largest) possible contamination bias from reordering the conditional ATEs to be as negatively (positively) correlated with the cross-treatment weights as possible. Panel B summarizes estimates of small class and teaching aide treatment effects from different specifications in the kindergarten sample of Project STAR. Column 1 reports estimates of small class and teaching aide treatment effects from the interacted model in eq. (28). Column 2 reports estimates from the treatment-specific regressions in eq. (29). Column 3 reports estimates from the efficiently weighted specification, using the estimated weights in eq. (30). Robust standard errors are reported in parentheses. Standard errors that assume the propensity scores are known are reported in square brackets.